FOREST ADVENTURE

Game Design Document

A small 3D adventure and collecting game for Windows PC

Unity 6 • Third-person exploration • Forest environment

1. Game Concept

Forest Adventure is a simple third-person 3D adventure and collecting game set in a stylized forest. The player explores the environment, collects five magical crystals and then finds a magical portal. Entering the portal after all crystals have been collected completes the game.

Genre: 3D adventure / exploration / collecting **Platform:** Windows PC **Engine:** Unity 6

Target audience: beginners and casual players who enjoy short, simple exploration experiences. The expected play session is approximately 5–10 minutes.

Design goal: demonstrate a complete, understandable gameplay loop without unnecessary complexity. The player should immediately understand what to do and receive clear feedback when progressing.

2. Gameplay

Core loop: Explore the forest → find crystals → collect all 5 → activate portal → enter portal → win.

Objective: collect five magical crystals and enter the portal that appears after the final crystal is collected.

Win condition: all five crystals collected and the activated portal entered. There is no traditional game-over state; the focus is exploration and completion.

3. Controls and User Interface

Input	Action
W / S	Move forward / backward
A / D	Move left / right
Mouse	Look around / control camera
ESC	Pause / resume

The main menu contains PLAY, HOW TO PLAY and QUIT. The How To Play screen explains the objective and controls. During gameplay, the top-left counter shows the current number of collected crystals. A pause menu provides RESUME and QUIT. After winning, the player can restart the game.

4. Visual Design

The visual style is a simple, colorful and readable stylized forest built from basic 3D primitives and simple materials. The ground is green and the environment contains trees, bushes, stones, tree stumps and colorful flowers. These decorative elements make the forest feel less empty while keeping the asset scope manageable.

Crystals: bright cyan/blue magical collectibles designed to stand out from the green environment. **Portal:** a magical visual element that becomes visible only after all crystals are collected. **UI:** clear text, buttons and dark forest-inspired panels.

5. Game Objects and Assets

Object / System	Purpose
Player	Third-person character using Character Controller.
Camera	Mouse-controlled third-person camera following the player.
Crystals	Five collectible objects required to finish the game.
Portal	Activates after all crystals and triggers the win state.
Trees, bushes, stones, stumps, flowers	Forest decoration and atmosphere.
Main Menu / Pause Menu	Navigation, instructions, pause, restart and quit.
Audio	Background music, crystal collection and portal sound.

6. Technical Design

The game is developed in Unity 6 with C#. Player movement uses a Character Controller. Trigger colliders handle crystal collection and portal interaction. The project separates functionality into scripts for player movement, camera control, collectibles, game management, portal behaviour, UI, crystal animation, pause management and main menu management. Unity UI and TextMeshPro are used for menus, counters and instructions. AudioSources provide background music and sound effects.

7. Project Scope and Future Development

The current scope is intentionally small but complete: exploration, collecting, UI, audio, pause functionality, main menu and a clear win condition. Possible future development includes a larger forest, multiple levels, more collectible types, environmental effects such as fog or fireflies, improved models and animations, puzzles or challenges, and a save/progression system.

8. Pre-production Decisions

The forest theme was selected because it supports a recognizable environment using simple assets and allows the project to remain manageable as a student game. The design prioritizes a complete player experience over technical complexity. Every major system supports the central objective of collecting crystals and reaching the portal.

Final target: a polished small Windows PC game that demonstrates the complete development process from design through implementation, testing and presentation.